Low-Cost Amphibious Robotic System for Freshwater Wetland Research and Exploration

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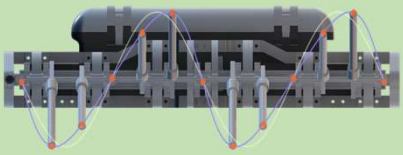
Motivation

- Wetlands are large carbon sinks, producing 20-30% of total annual global methane emissions
- Understanding wetlands better will improve our understanding of climate and the roles that this natural ecosystem plays within it
- Wetlands are notoriously difficult to navigate and traverse, posing as a barrier for research to be conducted in the influential habitats
- New robotic platforms allow for the exploration of such ecosystems, but these tend to be prohibitively expensive and inaccessible to researchers
- Our goal is to create an inexpensive, customizable amphibious robotic platform that would allow ecologists to map and monitor wetlands at a greater spatiotemporal scales than conventional field measurements

Challenges

- The robot will have to operate in a wide range of locations, where it is entirely possible that there will be no cellular service
- The robot must be designed to have maneuverability and stable motion through rugged terrain, dense growth, and water
- The robot must at least have an inner waterproof enclosure that protects the electronics from the environment, if not be entirely waterproof itself. It is challenging to make something that is both cost effective and waterproof
- To maintain a low barrier-of-entry, only general-purpose manufacturing equipment and 3D printer can be used to build and assemble the robot

Motion



The robot travels in an approximate sinusoidal pattern following a travelling wave. The legs extend and touch the ground with a minimum of two points of contact per side, providing the friction necessary to propel the robot forward. A silicon outer-skin will increase the traction our robot will need to locomote on land while provide the thrust to propel through water. The legs are driven by a single DC motor, with one on each side of the robot.

Waterproof Container



Waterproof box contains components such as the Raspberry Pi, two DC motors and motor controllers, RTK GNSS sensor, and two orthogonal IMUs. Epoxy resin will support print surfaces and allow waterproofing, along with seals for rotary shafts, box halves, lid, and wires.



Cameras



There are two cameras on the robot. A forward-facing camera allows for the robot to stream visual data to an operator's computer. A secondary camera is currently facing beneath the robot, providing visual information about what the robot is transversing over. This will eventually be moved to the top of the robot to provide periscope-like behavior, and can also be viewed by the operator.

This visual information allows the operator to safely maneuver the robot throughout the ecosystem without having to visually see the location of the robot.

Eventually, these cameras be used to perform obstacle avoidance while the robot navigates autonomously, as well as a providing a potential method for species recognition and detection.

Skin

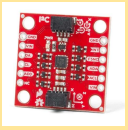


We are working on developing a waterproof, easy-to-assemble skin that wraps around the entire robot, providing a first layer of protection for the electronics and achieving propulsion in water.

We are considering TPU, which is printable and flexible, and silicon, which is may be harder to reproduce but would most likely increase the water resistance of the skin.

The skin will give our robot 'fins' to maneuver through the water, increase friction on surfaces when locomoting on land, and protect our robot from the harsh amphibious environments.

Sensors



There exist two IMUs within the robot, which are on orthogonal axes relative to each other. The IMUs will be used to estimate the pitch, roll, and yaw of the robot.



The ZED-F9P-05B is a high precision GNSS module with RTK, allowing the operator to send correctional data to the robot and making the accuracy to be on the order of centimeters.

Future work will increase the customizability of the sensor payload available to the robot. This may include moisture sensors, pH sensors, and oxygen saturation sensors. Additionally, we plan to add an auger module with which an operator will be able to dig into soil underneath the robot in to study its moisture and oxygen saturation levels.

Approach

- Based on the Velox platform developed by Pliant Robotics
- Waterproof box in the center of the chassis hosts all electronics, with a rotary seal interfacing between the motors and the gear shaft to prevent water leakage
- Long gear shaft approach replaces many individual servo motors with a singular DC motor, increases ease of control logic
- All electronics are off-the-shelf to further reduce cost and barrier to entry
- PLA chassis increases iterability
- Experimenting with both a silicon based mold for the fins, as well as potentially printing the fins out of TPU
- Separate rechargeable LiPo batteries for motor and RPi 5
- A high length-to-width ratio for the chassis will improve efficiency and speed when moving through water



Future Work

- Fully waterproof the electronics box with Epoxy
- Develop method for making the fins
- Test locomotion in water and on land with the fins
- Develop a customizable space within the robot with additional ecological sensors/actuators
- Redesign the chassis so that it is easier to build and requires less material
- Develop basic waypoint navigation and state-space control

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